

## Problem

Solve the following equations using matrix inversion.

$$\begin{aligned}y_1 - y_3 &= 1 \\2y_1 + 3y_2 - y_3 &= 1 \\y_1 - y_2 - y_3 &= 3\end{aligned}$$

## Step-by-step solution

## Step 1 of 11

Consider the following set of equations:

$$\begin{aligned}y_1 - y_3 &= 1 \\2y_1 + 3y_2 - y_3 &= 1 \\y_1 - y_2 - y_3 &= 3\end{aligned}$$

Arrange the equations in matrix form.

$$\underbrace{\begin{bmatrix} 1 & 0 & -1 \\ 2 & 3 & -1 \\ 1 & -1 & -1 \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}}_{\mathbf{Y}} = \underbrace{\begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}}_{\mathbf{B}}$$

The matrix equation is,

$$\mathbf{AY} = \mathbf{B}$$

Here,

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 3 & -1 \\ 1 & -1 & -1 \end{bmatrix}$$

$$\mathbf{B} = \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}$$

## Step 2 of 11

From the matrix equation,  $\mathbf{AY} = \mathbf{B}$ , system of equations are solved by invert the matrix  $\mathbf{A}$  and multiply it with  $\mathbf{B}$ .

$$\mathbf{Y} = \mathbf{A}^{-1}\mathbf{B}$$

Where,

$$\mathbf{A}^{-1} = \frac{\text{adj}(\mathbf{A})}{|\mathbf{A}|}$$

$$\text{adj}(\mathbf{A}) = (\mathbf{C})^T$$

Here,

$\mathbf{C}$  is the Cofactors of matrix  $\mathbf{A}$ .

$$\mathbf{C} = \text{cof}(\mathbf{A})$$

$$= \begin{bmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{bmatrix}$$

**Step 3** of 11

Calculate the co-factor  $c_{11}$ .

$$\begin{aligned} c_{11} &= \begin{vmatrix} 3 & -1 \\ -1 & -1 \end{vmatrix} \\ &= (3)(-1) - (-1)(-1) \\ &= -3 - 1 \\ &= -4 \end{aligned}$$

Calculate the co-factor  $c_{12}$ .

$$\begin{aligned} c_{12} &= - \begin{vmatrix} 2 & -1 \\ 1 & -1 \end{vmatrix} \\ &= -[(2)(-1) - (1)(-1)] \\ &= 2 - 1 \\ &= 1 \end{aligned}$$

**Step 4** of 11

Calculate the co-factor  $c_{13}$ .

$$\begin{aligned} c_{13} &= \begin{vmatrix} 2 & 3 \\ 1 & -1 \end{vmatrix} \\ &= (2)(-1) - (3)(1) \\ &= -2 - 3 \\ &= -5 \end{aligned}$$

Calculate the co-factor  $c_{21}$ .

$$\begin{aligned} c_{21} &= - \begin{vmatrix} 0 & -1 \\ -1 & -1 \end{vmatrix} \\ &= -[(0)(-1) - (-1)(-1)] \\ &= -[0 - 1] \\ &= 1 \end{aligned}$$

**Step 5** of 11

Calculate the co-factor  $c_{22}$ .

$$\begin{aligned} c_{22} &= \begin{vmatrix} 1 & -1 \\ 1 & -1 \end{vmatrix} \\ &= (1)(-1) - (1)(-1) \\ &= -1 + 1 \\ &= 0 \end{aligned}$$

**Step 6** of 11

Calculate the co-factor  $c_{23}$ .

$$\begin{aligned}c_{23} &= - \begin{vmatrix} 1 & 0 \\ 1 & -1 \end{vmatrix} \\&= -[(1)(-1) - (1)(0)] \\&= -[-1 - 0] \\&= 1\end{aligned}$$

Calculate the co-factor  $c_{31}$ .

$$\begin{aligned}c_{31} &= \begin{vmatrix} 0 & -1 \\ 3 & -1 \end{vmatrix} \\&= (0)(-1) - (3)(-1) \\&= 0 + 3 \\&= 3\end{aligned}$$

**Step 7** of 11

Calculate the co-factor  $c_{32}$ .

$$\begin{aligned}c_{32} &= - \begin{vmatrix} 1 & -1 \\ 2 & -1 \end{vmatrix} \\&= -[(1)(-1) - (2)(-1)] \\&= -[-1 + 2] \\&= -1\end{aligned}$$

**Step 8** of 11

Calculate the co-factor  $c_{33}$ .

$$\begin{aligned}c_{33} &= \begin{vmatrix} 1 & 0 \\ 2 & 3 \end{vmatrix} \\&= (1)(3) - (2)(0) \\&= 3 - 0 \\&= 3\end{aligned}$$

Therefore, the cofactors of A are,

$$C = \begin{bmatrix} -4 & 1 & -5 \\ 1 & 0 & 1 \\ 3 & -1 & 3 \end{bmatrix}$$

Calculate the adjoint of matrix **A**.

$$\begin{aligned}\text{adj } \mathbf{A} &= \begin{bmatrix} -4 & 1 & -5 \\ 1 & 0 & 1 \\ 3 & -1 & 3 \end{bmatrix}^T \\&= \begin{bmatrix} -4 & 1 & 3 \\ 1 & 0 & -1 \\ -5 & 1 & 3 \end{bmatrix}\end{aligned}$$

**Step 9** of 11

Calculate the determinant of matrix **A**.

$$\begin{aligned} |\mathbf{A}| &= (1)c_{11} + (0)c_{12} - (1)c_{13} \\ &= (1)(-4) + 0 - (1)(-5) \\ &= -4 + 0 + 5 \\ &= 1 \end{aligned}$$

Determine the inverse of **A**.

$$\begin{aligned} \mathbf{A}^{-1} &= \frac{\text{adj}(\mathbf{A})}{|\mathbf{A}|} \\ &= \frac{1}{1} \begin{bmatrix} -4 & 1 & 3 \\ 1 & 0 & -1 \\ -5 & 1 & 3 \end{bmatrix} \\ &= \begin{bmatrix} -4 & 1 & 3 \\ 1 & 0 & -1 \\ -5 & 1 & 3 \end{bmatrix} \end{aligned}$$

**Step 10** of 11

Calculate the system matrix **Y** .

$$\begin{aligned} \mathbf{Y} &= \mathbf{A}^{-1}\mathbf{B} \\ \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} &= \begin{bmatrix} -4 & 1 & 3 \\ 1 & 0 & -1 \\ -5 & 1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix} \\ &= \begin{bmatrix} (-4)(1) + (1)(1) + (3)(3) \\ (1)(1) + (0)(1) + (-1)(3) \\ (-5)(1) + (1)(1) + (3)(3) \end{bmatrix} \\ &= \begin{bmatrix} -4 + 1 + 9 \\ 1 + 0 - 3 \\ -5 + 1 + 9 \end{bmatrix} \\ &= \begin{bmatrix} 6 \\ -2 \\ 5 \end{bmatrix} \end{aligned}$$

**Step 11** of 11

Therefore, the solution to the simultaneous equations is,

$$\begin{bmatrix} y_1 = 6 \\ y_2 = -2 \\ y_3 = 5 \end{bmatrix}.$$